

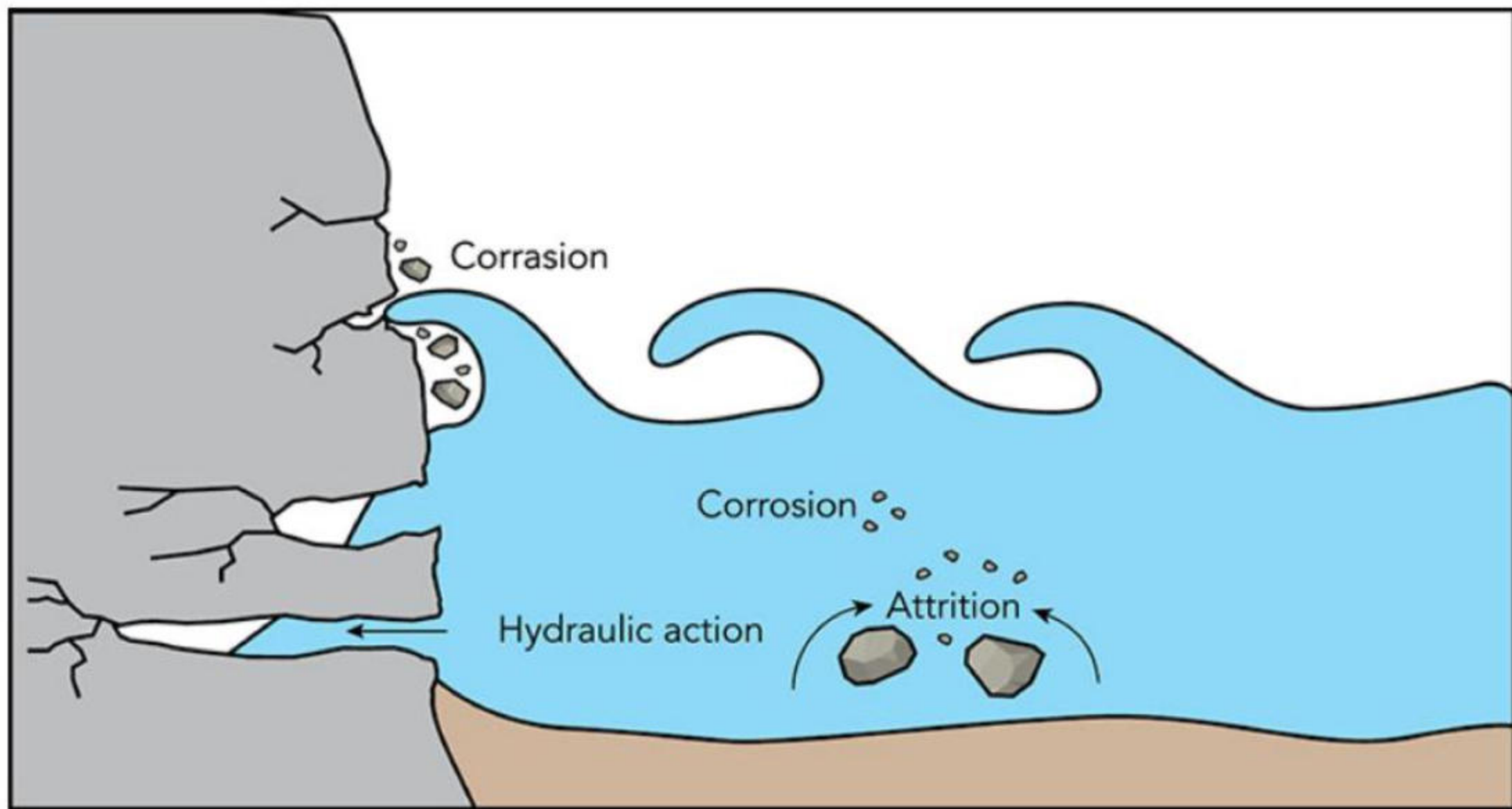
2.1 How physical processes shape the coast

Like rivers, coasts are dynamic environments that are constantly changing. Here, processes of erosion, deposition, and transportation are at work. These processes help shape the landscape and landforms you find along the coast.

There are different types of erosion at work in coastal environments. They are hydraulic action, **corrasion**, attrition, and **corrosion**. These processes break down the land. They break down cliffs or material that is already in the sea, such as pebbles, making them smaller and smoother over time.

TIP

These erosional processes work in the same way as they do in rivers. Just remember to refer to the correct environment (rivers or coasts) in your answers.



Source **A**: Coastal erosion processes

Once material has been eroded into the sea, the sea then moves or transports it. This transportation occurs as a result of winds, tides, and currents. The size and weight of the material determines how it moves. For example, as with rivers, larger, heavier particles move by traction, whereas the lightest material is carried in suspension. Particles that are too light to roll along the seabed move by saltation in a series of hops. Some material dissolves in the sea and then moves in solution.

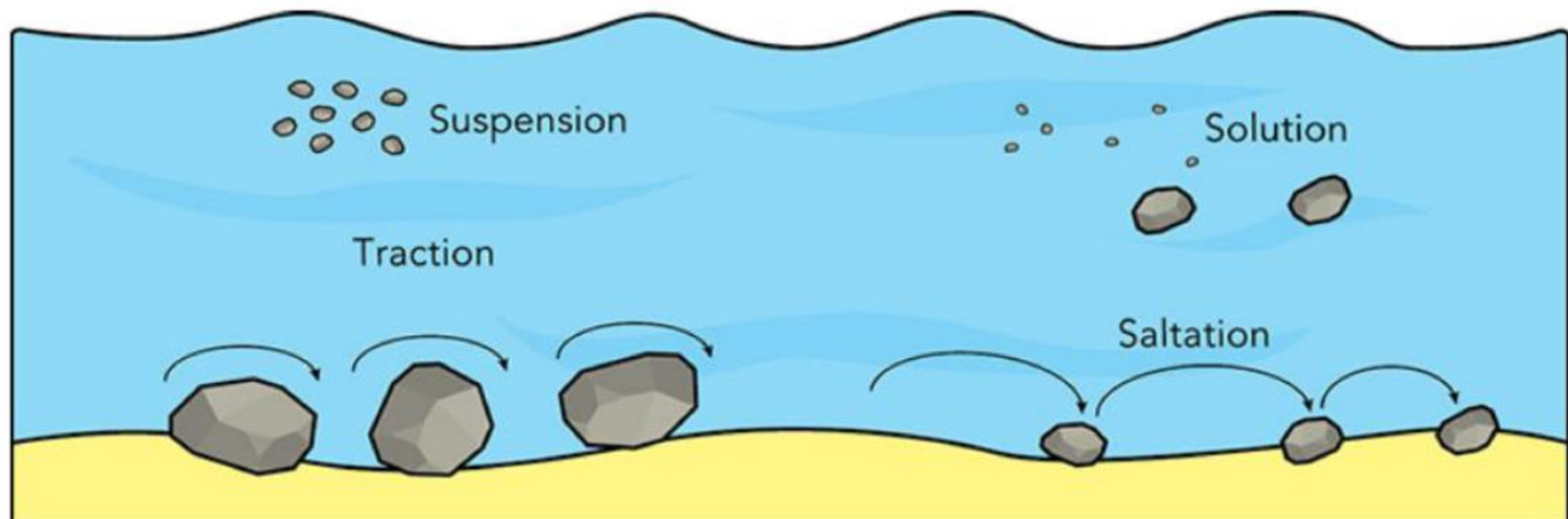


Figure 2.1: Coastal transportation processes

TASK 1: Study Source **A**

- a** Draw a table with the headings shown in the table below.
Complete the table for the following key terms:

Key term	Definition	Erosion or transportation	Diagram
Saltation			

b Identify the processes that are missing and add them to your table.

A transportation process called **longshore drift** determines the overall direction of material moving along the coast. The prevailing (most common) wind direction determines which way sediment is moved along a coastline. The prevailing wind blows waves onto the shore at an angle. This water is the **swash**, and it pushes material up onto the beach. The **backwash** (water from a wave that moves back to the sea) pulls material back down the beach at right angles to the coast due to gravity. This 'zig-zag' process repeats over time. It results in material moving further along the coast in the direction of the longshore drift.

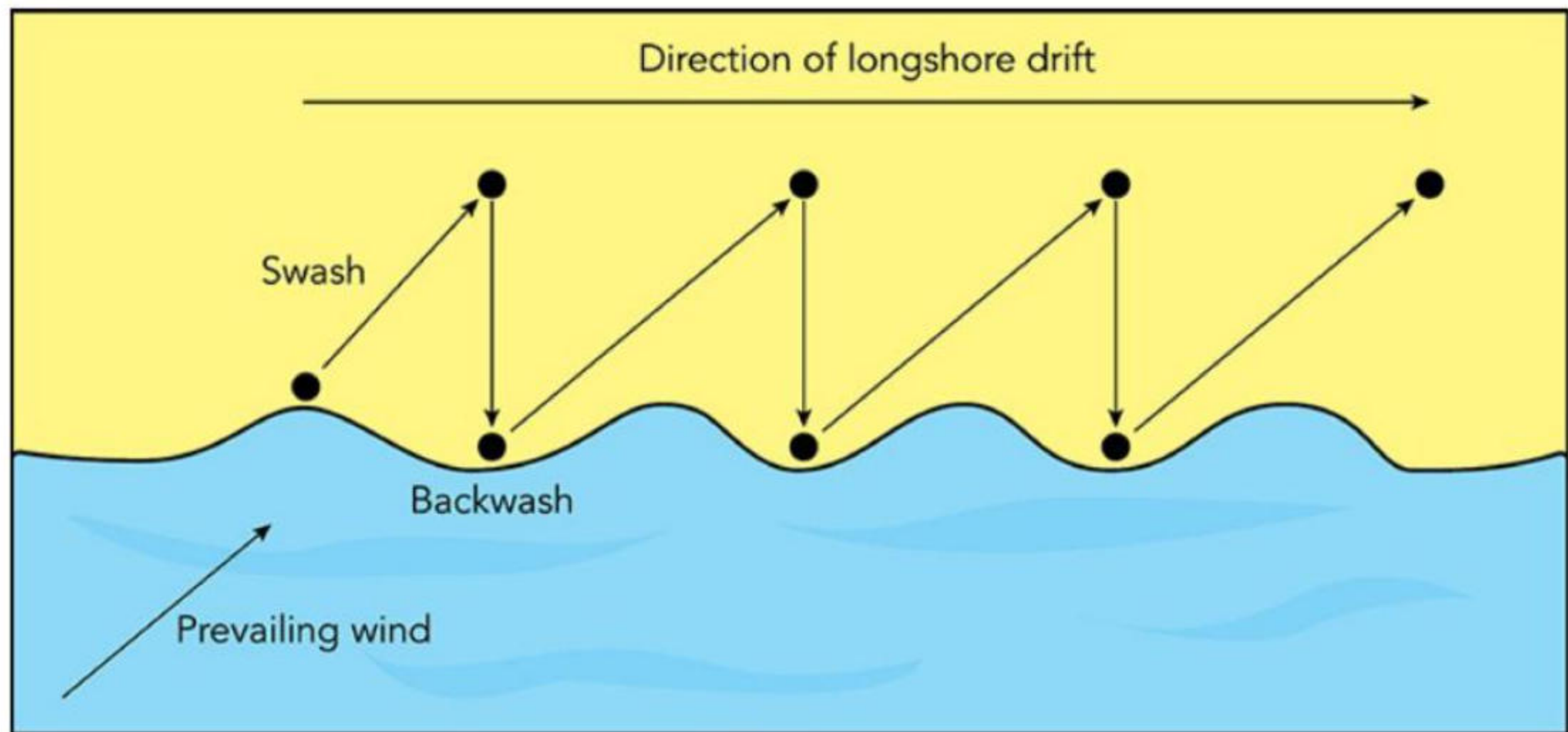
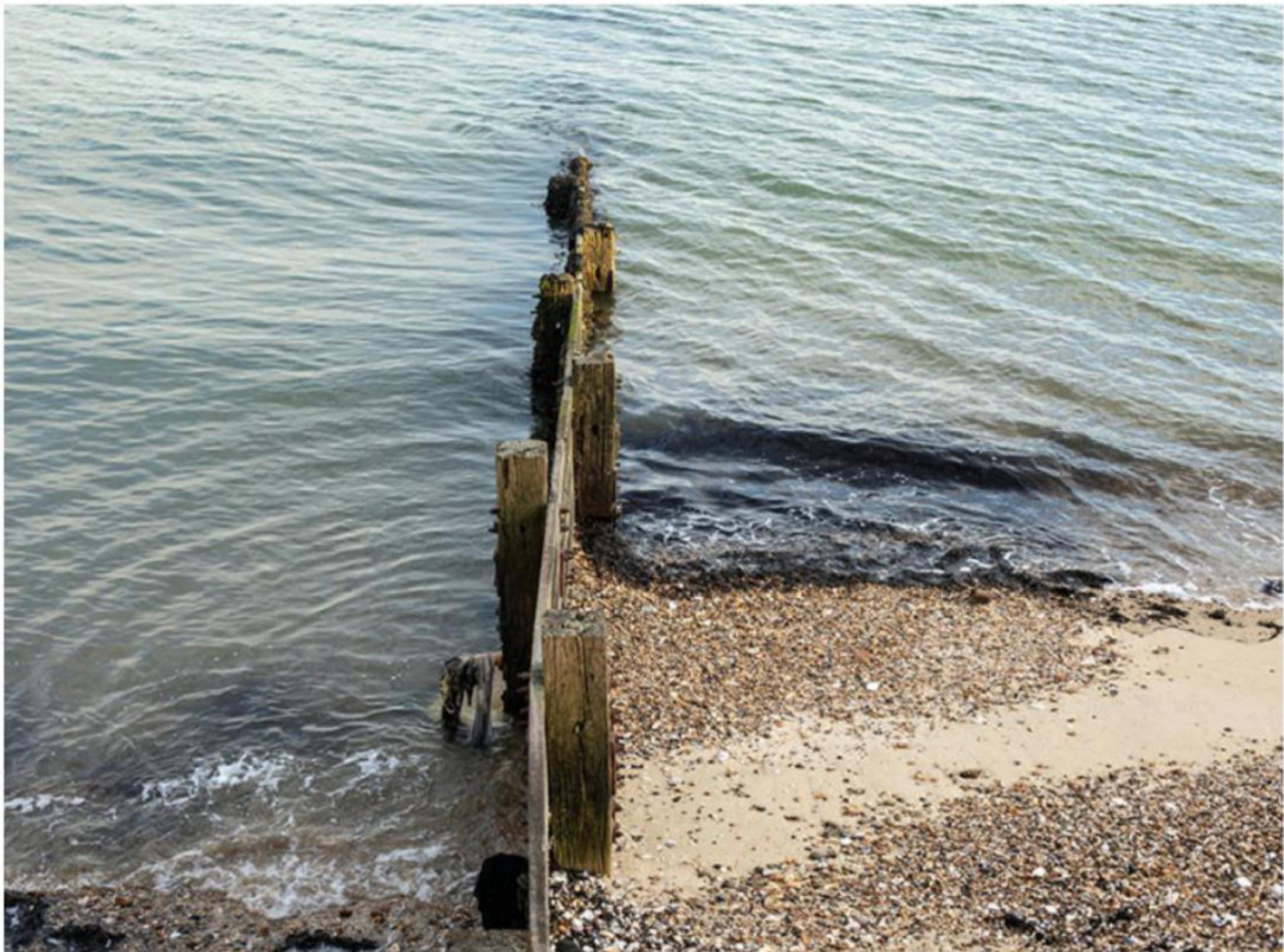


Figure 2.2: Longshore drift



Source B: A wooden groyne. Groynes trap sediment that is transported along the coast by longshore drift.

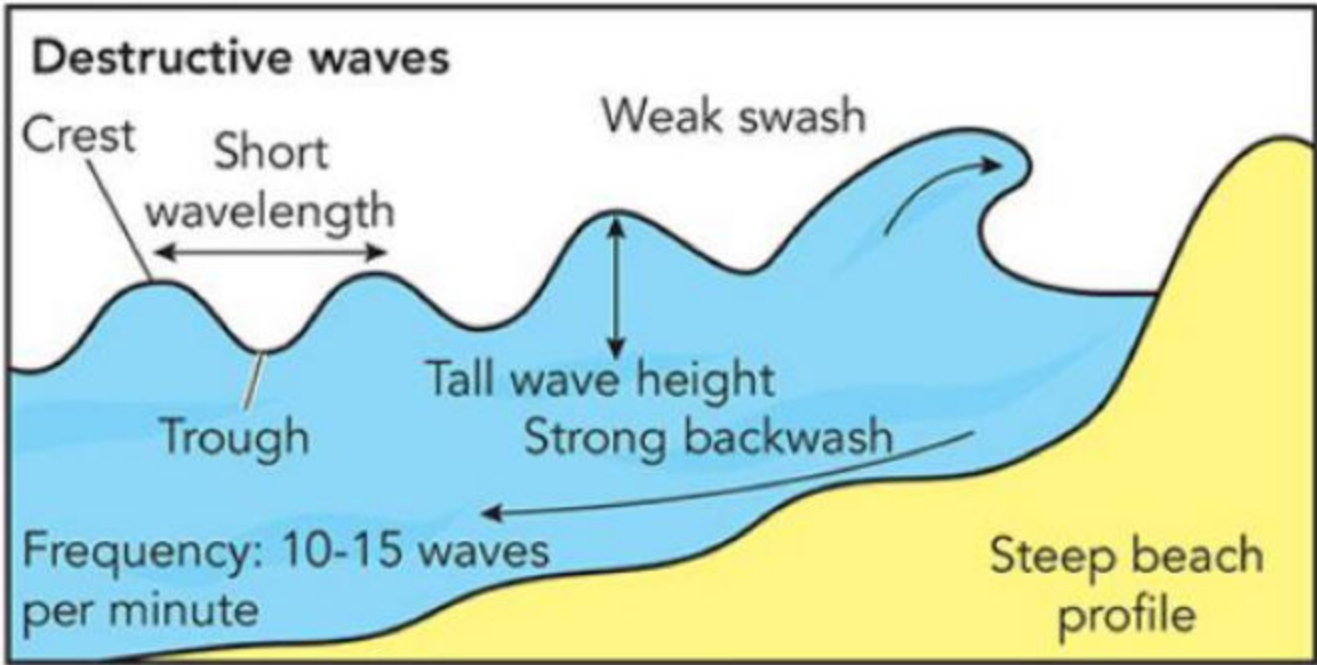
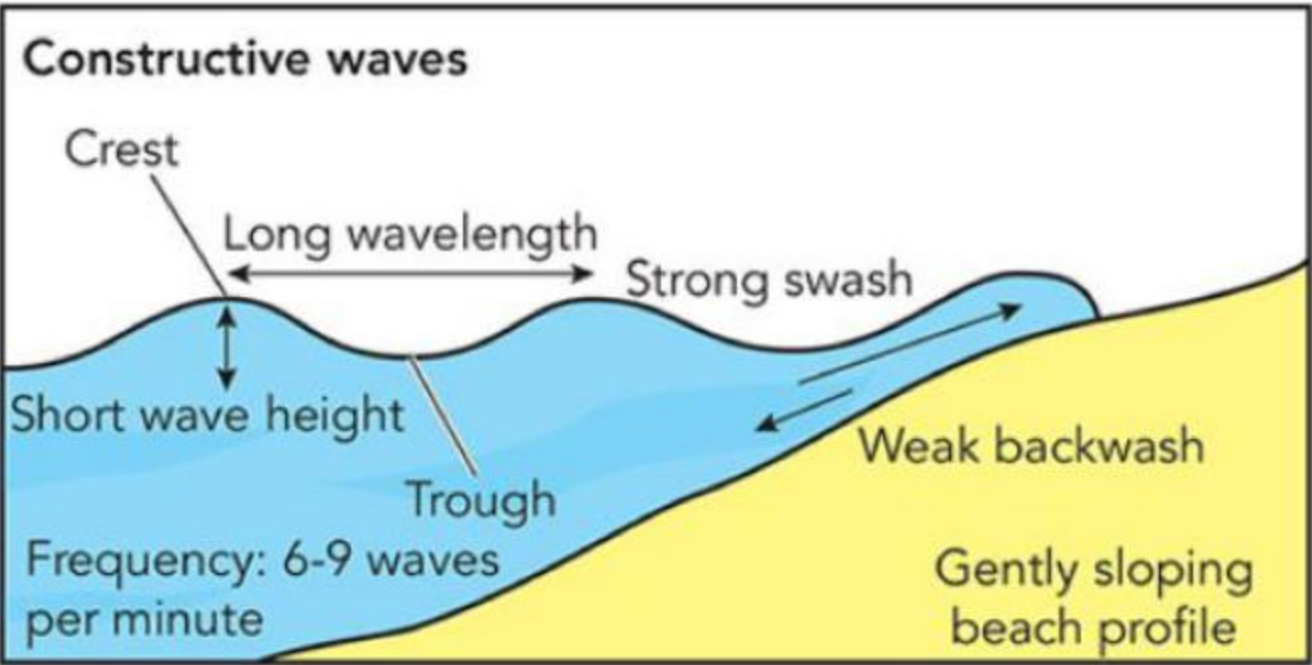
TASK 2: Study Source B

- a What evidence can you see for the process of longshore drift?
- b What is the direction of longshore drift here?
- c What do you think the impact of installing groynes here will have further along the coast?

Waves

Waves are caused by wind blowing over the sea. The **fetch** (the distance of the sea that the wind blows over) determines the power of the waves hitting the coastline. A shorter fetch produces less powerful waves than a longer fetch, which produces very powerful waves. This is because the wind can blow for longer over a greater distance and build up more energy.

All waves have a crest, which is the highest point of the wave, and a trough, which is the lowest point of the wave. There are two types of wave – **constructive** and **destructive**. Constructive waves have a long **wavelength**, low **wave height**, and low **wave frequency**. They also have a stronger swash than backwash. This produces gently sloping beaches as material piles up through deposition. Destructive waves have a short wavelength, high wave height, and high wave frequency. They have a stronger backwash than swash. This erodes the beach, resulting in a steep beach profile.



Source C: Constructive and destructive waves

TASK 3: Study Source C

Describe the properties of each wave type (constructive and destructive). Refer to wave characteristics, such as wavelength and amount of swash. Test yourself with a partner.

TASK 4

- a Using Tables 2.1 and 2.2, draw two scatter graphs that show the profile of each beach. show the profile of each beach.

Table 2.1: Beach A

Distance from the sea (m)	Beach height (m)
0	0
10	0.6
20	1.4
30	1.2
40	1.3
50	1.2

- b Decide which beach is formed by constructive waves and which is formed by destructive waves. Give reasons for your answers.

Table 2.2: Beach B

Distance from the sea (m)	Beach height (m)
0	0

10	0.4
20	0.7
30	0.9
40	1.1
50	1.4



Figure 2.3: A gentle beach profile created by constructive waves



Figure 2.4: A steep beach profile created by destructive waves

Wave refraction occurs when waves approach a coastline with varying depths. The varying depths are due to the creation of headlands and bays over time. As waves travel into shallower water, they slow down. This is due to the decrease in water depth and increase in friction. Shallower water is found at headlands before bays, which results in the wave slowing down more here than the part of the wave in deeper water. This causes the wave to bend or refract. The wave energy is focused on headlands, causing more erosion, rather than in bays where deposition of sediment occurs.

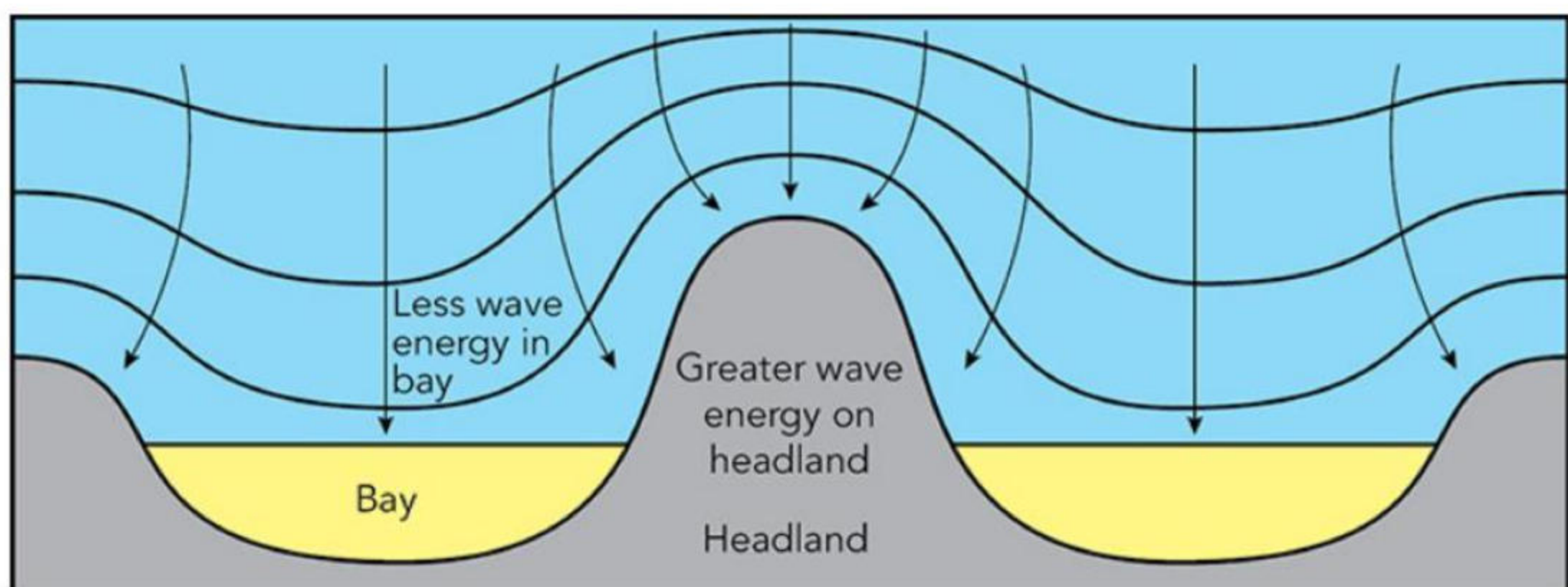


Figure 2.5: Wave refraction